



**UNIVERSITY
OF LONDON**

CM3070 Preliminary Project Report

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**ARIA – Activity Recognition & Inference Architecture, A
Multi-Person Activity Detection and Role-Based Tracking
System in a Shared Service Counter Environment**

**Template: CM3065 13.2 – Multi-human Detection and
Activities Tracking in a Shared Space**

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1. Introduction

1.1. Project Idea

This project proposes the development of ARIA (Activity Recognition & Inference Architecture), a privacy-preserving, multimodal system that detects and tracks the activities of multiple staff members operating in a shared ticketing counter environment. The system identifies who is performing which activity at any given time, using a combination of laptop-based computer vision, ambient audio analysis, and supplementary IoT sensing via an Arduino ESP8266 node.

The target scenario features two distinct staff roles coexisting in one shared physical space: counter agents (stationed at the front, serving customers) and station managers (moving between the front and back office, supervising operations and handling administrative tasks). The system tracks role-specific activities in real time and produces a labelled, publishable dataset as a secondary contribution.

1.2. Motivation

1.2.1. *The Problem*

Shared service environments, such as transport ticketing counters, bank branches, and service desks, routinely involve multiple staff members with different roles performing different activities simultaneously in a single space. Operational decision-making in such environments, such as when to deploy additional counter agents, or when a manager's intervention is needed, currently relies on manual supervision or post-hoc reviews of CCTV footage. Neither approach is efficient, automated, or scalable.

Ambient Assisted Living (AAL) research has made significant progress in detecting activities of daily living, but existing systems are largely designed for single-occupant or undifferentiated multi-occupant settings. They cannot accurately attribute specific activities to specific individuals when multiple people share a space, nor do they address the domain-specific needs of a workplace service counter context.

A further challenge is the privacy and cost burden of cloud-based video processing. Streaming live video to cloud services raises significant data protection concerns, consumes substantial bandwidth, and incurs ongoing financial costs, all of which are identified as open challenges in the brief's background context.

1.2.2. *Why This Project Matters*

ARIA addresses a practical operational challenge in shared service environments. Managers need real-time, role-specific activity attribution (e.g., serving, waiting, moving) rather than simple occupancy counts, enabling evidence-based operational decision-making.

Technically, ARIA tackles multi-user activity recognition, a harder problem than single-user HAR. It maintains identity continuity and links activities to specific individuals in shared workspaces despite

occlusion and overlapping movement. ARIA demonstrates that this complex attribution can be achieved using low-cost hardware rather than specialised lab infrastructure.

Furthermore, ARIA prioritises privacy by design. Instead of continuous cloud video processing—which raises bandwidth, cost, and data-protection concerns—ARIA performs local edge inference and only transmits abstracted feature data.

Finally, ARIA contributes a publishable, labelled multimodal dataset for counter-staff activities. This underrepresented domain dataset will support ARIA’s evaluation and provide a baseline for future workplace-oriented, multi-subject HAR research.

1.3. Project Concept

1.3.1. System Overview

ARIA operates across the following three tiers:

1.3.1.1. Tier 1 – Edge Sensor Node (Arduino ESP8266)

Three PIR motion sensors and two ultrasonic distance sensors are deployed across two named zones (counter zone, back zone), with an analogue microphone module on A0. The ESP8266 reads all sensors and transmits JSON payloads over serial USB to the edge inference host.

1.3.1.2. Tier 2 – Edge Inference Host (Laptop/PC)

A Python application subscribes to the ESP8266 serial stream and simultaneously captures webcam video and microphone audio. A YOLOv8n model detects and tracks persons (ByteTrack); a pose estimation model (MediaPipe or YOLOv8-pose) extracts skeletal keypoints. Combined with sensor features, a 1D-CNN or Random Forest classifier identifies the activity class of each tracked individual. Results are logged to a timestamped CSV dataset and displayed on a real-time Dash dashboard.

1.3.1.3. Tier 3 – Cloud Layer (Google Colab + Firebase)

Feature vectors, not raw video, are periodically synced to Firebase Realtime Database. Google Colab retrains the classifier on accumulated data and pushes updated model weights back to the inference host in an iterative loop.

1.3.2. Activity Classes

Table 1: Activity Classes

Role	Activity Classes
Counter Agent	Serving (facing forward, leaning in), typing/processing (keyboard interaction), idle/waiting, reaching/handling
Station Manager	Walking (zone transition), desk work (seated, screen/paperwork), in-conversation (facing another staff member)

1.3.3. Hardware

- Laptop/PC – edge inference host (existing hardware)
- Arduino ESP8266 NodeMCU – IoT Sensor Hub
- 3 x HC-SR501 PIR Sensor – motion detection per zone
- 2 x HC-SR04 Ultrasonic Sensor – proximity/zone confirmation
- MAX9814 Microphone Module – ambient audio amplitude
- USB Webcams

1.4. Proposed Outcomes

1.4.1. Minimum (Pass)

A prototype running on a PC/edge device that can process vision and audio signals to detect and track at least two activity types across two simultaneous participants.

1.4.2. Good (Merit)

A hybrid edge-cloud prototype demonstrating the full training-inference-retraining loop, accurately detecting at least four activity types. A sufficiently large annotated dataset with documentation.

1.4.3. Outstanding

As above, plus a robustly validated system with per-class F1 metrics, ablation study results, and a publishable labelled dataset with full data card documentation, produced with the written consent of participants and suitable for submission to a public data repository like Kaggle or Zenodo.

2. Literature Review

This literature review evaluates seven bodies of work that collectively ground ARIA's design: studies validating multimodal sensing and edge inference, a collaborative edge-cloud architecture for multi-user recognition, a healthcare survey justifying the audio-vision fusion strategy, a dataset construction methodology, a multi-object tracking algorithm, and a survey identifying the multi-subject attribution gap that ARIA targets. Rather than summarising these works in isolation, the review draws cross-cutting connections, highlights convergences, identifies conflicting constraints, and maps each source to specific design decisions within ARIA.

2.1. Multimodal Sensing and Edge Inference

2.1.1. Nakabayashi & Saito (2024) – Multimodal HAR on Edge Devices

Nakabayashi and Saito demonstrated that fusing camera images with IMU data (acceleration and angular velocity) within a lightweight attention-mechanism network achieves approximately 97% precision for classifying five activity classes on the UESTC-MMEA-CL dataset, subsequently outperforming single-sensor baselines. Critically, the network was designed specifically for edge devices with limited computational resources, proving that high-accuracy multimodal inference does not require cloud offloading.

This work is directly foundational to ARIA's architecture. It validates two of ARIA's core bets: first, that fusing heterogeneous sensor streams (vision, audio, IoT) outperforms any single modality, and secondly, that a compact, efficient model can perform this fusion locally without server-side compute. The attention mechanism that dynamically weights sensor contributions depending on input quality is also conceptually aligned with ARIA's feature-vector fusion, where PIR, ultrasonic, audio amplitude, and skeleton keypoint features are combined before classification.

However, the single-user scope of Nakabayashi and Saito represents its most significant limitation relative to ARIA's requirements. The paper does not address identity disambiguation when two or more people share the frame, nor does it consider role-differentiated attribution. This gap – that high-performing single-user multimodal HAR breaks down under multi-occupancy – is precisely what ARIA is designed to address.

2.1.2. Golipoor & Sigg (2024) – RFID-Based HAR Using Multimodal CNNs

Golipoor and Sigg introduced a model for human activity recognition that leverages battery-less, body-worn reflective antenna components combined with multimodal convolutional neural networks, targeting industrial and factory automation settings where safety, efficiency, and productivity monitoring are important. Their work demonstrated that non-visual passive sensing can be combined with neural classification pipelines to recognise activities in professional environments, a setting closer to ARIA's service counter domain than the healthcare or domestic contexts that dominate most HAR literature.

This paper is relevant to ARIA on two levels. First, it validates the use of passive, ambient sensors, rather

than cameras alone, as a viable strategy for workplace activity recognition, providing precedent for ARIA's hybrid sensing approach combining PIR, ultrasonic, and audio modules. Second, it establishes that CNN-based classifiers are effective at processing heterogeneous, non-visual sensor inputs, supporting ARIA's choice of a 1D-CNN as one of its candidate classifiers.

The key contrast with ARIA lies in sensing modality: RFID requires wearable tags attached to participants, introducing a cooperation dependency. ARIA's design choice of entirely passive ambient sensing addresses this directly, making it deployable in naturalistic environments where workers are not expected to carry or wear tracking hardware.

2.2. Multi-User Activity Recognition Architectures

2.2.1. Li, Gravina, Savaglio & Fortino (2023) – BSN-Enabled Collaborative Edge-Cloud Architecture

Li et al. proposed a hybrid architecture in which body sensor networks (BSNs) perform local inference at the edge while model training is offloaded to the cloud, enabling multi-user activity recognition across distributed IoT participants. The paper demonstrated that separating the training phase (resource-intensive and cloud-hosted) from the inference phase (lightweight and edge-hosted) is both feasible and effective, achieving multi-user recognition without requiring all computation to occur at a central server.

This directly validates ARIA's three-tier architecture. ARIA's Tier 2 (laptop edge inference) mirrors the BSN inference layer, while ARIA's Tier 3 (Google Colab retraining + Firebase sync) mirrors the cloud training layer. The edge-cloud split is not merely a technical convenience in ARIA; it is a privacy-by-design decision.

By transmitting only statistical feature vectors to the cloud rather than raw video, ARIA follows the same principle Li et al. established: meaningful inference can be performed locally, with only abstracted representations sent upstream.

The critical limitation of Li et al. is the wearable BSN dependency. Body sensor networks require participants to actively wear sensor hardware throughout the monitoring period, which is impractical for a passive service counter environment where staff members should not be required to carry monitoring devices. ARIA substitutes wearable sensors with ambient sensing (PIR, ultrasonic, microphone, webcam), making the architecture passive and non-intrusive. The edge-cloud paradigm is preserved; the sensing layer is reimagined.

2.2.2. Arrotta, Civitarese, Chen, Cumin & Bettini (2025) – Multi-Subject HAR Survey

Arrotta et al. published a comprehensive survey of multi-subject human activity recognition, formalising the problem and cataloguing its open challenges. The survey identifies that while single-subject HAR has achieved mature performance across many sensing modalities, multi-subject HAR remains an open and

complex problem, particularly in scenarios where multiple occupants are engaged in simultaneous, potentially overlapping activities. The authors note a persistent shortage of multi-subject datasets, underdeveloped evaluation protocols for per-subject attribution, and the particular difficulty of activity disambiguation in shared spaces.

This survey performs a dual function for ARIA. On one hand, it provides authoritative confirmation that ARIA's target problem, role-differentiated activity attribution in a shared space, is indeed an acknowledged unsolved extension of the existing HAR literature, not a problem already addressed elsewhere. On the other hand, it prescribes evaluation standards: per-class F1 scoring, confusion matrices, and leave-one-session-out cross-validation are the expected evaluation protocols for any credible multi-subject HAR system.

Arrotta et al. also underscore the value of new datasets. The survey highlights the scarcity of multi-subject annotated datasets as a major bottleneck for progress in the field. ARIA's secondary contribution, a labelled, publishable dataset of counter-staff activities in a shared space, directly responds to this identified gap.

2.3. Multimodal Fusion Strategy: Justification for Audio & Video

2.3.1. Cristina, Despotovic, Pérez-Rodríguez & Aleksic (2024) – Audio- and Video-Based HAR Systems in Healthcare

Cristina et al. conducted a scoping review of audio- and video-based activity recognition systems in healthcare, covering complexity, performance, robustness, scalability, and privacy considerations. A central finding is that combining audio and video modalities significantly improves HAR accuracy over single-modality systems, particularly for activities with similar visual signatures. For instance, distinguishing seated-desk-work from seated-phone-conversations, where the visual difference is subtle but the audio signature differs substantially.

For ARIA, this paper provides systematic evidence for the audio-vision multimodal design. The activity taxonomy ARIA targets includes precisely such visually ambiguous pairs: a counter agent typing versus a counter agent speaking on the phone, or a station manager performing desk work versus a station manager in active conversation with a colleague. The audio channel, captured via a MAX9814 microphone module, provides discriminative signal for these cases that the skeletal keypoint stream alone cannot reliably separate.

However, the healthcare framing of Cristina et al. introduces an important constraint: their reviewed systems operate in single-patient or clearly separated multi-patient clinical environments, not shared occupancy spaces. Privacy and scalability findings from healthcare may not transfer directly to a workplace counter context. ARIA's architecture reflects this distinction by adopting ROI masking and operating entirely at the edge, measures not required in most clinical deployments but essential when cameras face towards staff in a semi-public workspace.

2.4. Dataset Construction Methodology

2.4.1. Moreaux, Ortiz, Ferrané & Lerasle (2019) – Kitchen20 Dataset for Audio-Based Action Recognition

Moreaux et al. introduced the Kitchen20 dataset, a raw-audio, environmental, kitchen-related, non-vocal dataset designed to fill a gap in domain-specific audio activity data. The paper contributed not only the dataset itself, but a replicable methodology: structured activity scripting, per-class audio segmentation, human annotation validation, and benchmark comparisons with established datasets such as ESC-50. The authors also proposed merging Kitchen20 with ESC-50 to form a larger ESC-70 benchmark, demonstrating how domain-specific datasets can complement general-purpose ones.

Kitchen20 is the most methodologically transferable work to ARIA's dataset contribution. ARIA's planned dataset, labelled multimodal sensor recordings of counter-staff activities, collected in a controlled simulated environment, will follow the same core protocol: scripted activity sequences, per-class labelling, and documentation suitable for public repository submission. The specific parallels include activity scripting to ensure class coverage, timestamped annotation to align multimodal sensor streams, and data card documentation describing collection conditions, participant demographics, and sensor configurations. The key difference is modality depth: Kitchen20 is audio-only and single-person, while ARIA's dataset will be multimodal (vision keypoints, audio amplitude, PIR/ultrasonic readings) and multi-person. Kitchen20 illustrates what a well-constructed domain-specific audio dataset looks like. ARIA expands this template to a richer sensor stack and a more complex multi-occupant labelling problem.

2.5. Core Algorithmic Components

2.5.1. ByteTrack – Multi-Object Tracking by Associating Every Detection Box (Zhang et al., 2022)

ByteTrack, published at ECCV 2022, addresses a fundamental limitation in multi-object tracking: prior methods discard low-confidence detection boxes, which frequently correspond to occluded or partially visible people, generating fragmented trajectories. ByteTrack's contribution is to associate every detection box, including low-score ones, with existing tracklets using motion similarity, achieving 80.3 MOTA, 77.3 IDF1, and 63.1 HOTA on the MOT17 benchmark at 30 frames per second on a single V100 GPU.

ByteTrack is ARIA's chosen multi-person tracking backbone, integrated via the Ultralytics YOLOv8 tracking API. Its relevance is direct: in the service counter environment, a staff member may be partially occluded by a counter fixture, another person, or the camera angle. Without ByteTrack's low-score association mechanism, such a person would produce a fragmented track, breaking the identity continuity required for per-individual activity attribution. ByteTrack's real-time performance at 30 FPS also ensures that the tracking layer does not become a bottleneck in ARIA's pipeline.

ByteTrack does not, on its own, perform activity classification; rather, it provides stable, consistent identities across frames. The skeleton keypoint extraction (MediaPipe / YOLOv8-pose) and the downstream classifier then operate on these stable track IDs to assign activity labels. This layered separation of tracking from classification is a deliberate architectural choice that reflects ByteTrack’s design philosophy: reliable tracking as an enabler, not a classifier.

2.6. Synthesis and Critical Comparison

The seven works reviewed span four functional dimensions of ARIA:

- 1. multimodal sensing and edge inference,
- 2. multi-user architecture,
- 3. fusion strategy justification, and
- 4. algorithmic components and dataset methodology.

The table below maps each source to its contribution and the gap it highlights.

Table 2: Core Contributions and Gaps

Source	Domain	Core Contribution to ARIA	Gap ARIA Addresses
Nakabayashi & Saito (2024)	HAR / Edge AI	Validates multimodal fusion + edge inference; ~97% precision	Single-user only; no attribution in shared space
Li et al. (2023)	IoT / Edge-Cloud	Validates edge-cloud split for multi-user HAR	Wearable sensor dependency; not passive
Cristina et al. (2024)	Healthcare HAR	Audio + video fusion outperforms single modality	Healthcare framing; no multi-person attribution
Moreaux et al. (2019)	Audio datasets	Domain-specific dataset construction methodology	Audio-only; single-person; no multimodal fusion
Arrotta et al. (2025)	Multi-subject HAR	Identifies multi-subject attribution as an open problem	Dataset scarcity; no workplace domain coverage
Zhang et al. / ByteTrack (2022)	Computer Vision	Robust real-time multi-person identity tracking	No activity classification; tracking only
Golipoor & Sigg (2024)	Industrial HAR	Passive sensor + CNN for workplace activity recognition	Wearable RFID tags required; not ambient

A recurring theme across the reviewed literature is the unsolved transition from single- to multi-person recognition. While Nakabayashi & Saito, Cristina et al., and Moreaux et al. focus on single-occupant scenarios, Li et al. address multi-user HAR but rely on wearable BSNs. As Arrotta et al. confirm, passive, ambient multi-subject attribution in shared spaces remains a frontier challenge.

Secondly, the tension between cloud processing and privacy is evident. Both Li et al. and Cristina et al. highlight data protection concerns with cloud video processing. ARIA implements privacy-by-design

principles through local edge inference, ROI masking, and feature-vector-only transmission, satisfying GDPR requirements for semi-public spaces.

Thirdly, there is a trade-off between modality intrusiveness and deployment practicality. Wearables offer high accuracy but require active user cooperation. ARIA opts for passive ambient sensing to ensure naturalistic deployment, using ByteTrack to maintain persistent visual identities and overcome the inherent ambiguity of non-wearable tracking. Finally, ARIA's secondary dataset contribution will help address the critical shortage of multi-subject, workplace-oriented HAR benchmarks.

3. Design

3.1. Domain and Users

3.1.1. Domain

ARIA operates in the domain of Human Activity Recognition (HAR), specifically at the intersection of multi-subject recognition, edge AI inference, and workplace operational monitoring. It is designed for a shared service counter environment in which two staff roles perform simultaneous, potentially overlapping activities in the same space.

3.1.2. Users

The system is designed for two primary user groups, counter agents and station managers, whose tracked activity classes are defined in Table 1.

Although the immediate human subjects in the environment are counter agents and station managers, the wider beneficiary of the system is the operational decision-maker responsible for maintaining service efficiency. For counter agents, the system's value lies indirectly in making workload distribution visible and in distinguishing active service from idle or waiting states without requiring self-reporting or wearable devices. For station managers, the system provides a structured view of how supervisory movement, desk work, and interaction time are distributed across a shift. More broadly, separating these two roles is essential because the same movement pattern may carry different operational meaning depending on who performs it; for example, walking may indicate supervision for a manager but task transition or queue handling for another staff role.

3.2. Design Justification

3.2.1. Multimodal Sensing

ARIA fuses computer vision, ambient audio, and IoT sensing because prior work shows multimodal fusion improves robustness for visually ambiguous activities while remaining feasible on edge hardware. Vision provides spatial and postural data. Audio adds contextual cues for activities with similar upper-body postures (e.g., typing vs. phone calls). IoT signals (PIR/ultrasonic) provide environmental context like zone occupancy. Together, they complement each other, reducing single-stream failure risks.

3.2.2. Edge-First, Privacy-by-Design Architecture

All vision and ML processing runs locally on the inference host (laptop), with only statistical feature vectors, never raw video, transmitted to the cloud. As established in the literature synthesis, this edge-first architecture is a regulatory necessity for preserving privacy in semi-public spaces, separating lightweight local inference from resource-intensive cloud retraining.

The edge-first design is also justified by the practical deployment characteristics of the target environment. A shared service counter is a semi-public workspace in which staff may be continuously visible to cameras while customers may appear transiently at the boundary of the scene, making indiscriminate cloud transmission of raw footage undesirable from both an ethical and operational perspective. Local inference reduces dependency on stable upstream connectivity, lowers recurring bandwidth requirements, and makes the system more resilient in environments where network reliability cannot be guaranteed throughout the day. The cloud layer is retained only for the part of the workflow that genuinely benefits from centralisation,

namely periodic retraining on accumulated feature data, so the design preserves adaptability without sacrificing the principle that raw visual data should remain local wherever possible.

3.2.3. Passive Ambient Sensing over Wearables

ARIA deliberately avoids wearable sensors because they require active participant cooperation, which is impractical in a naturalistic service counter environment. Instead, ByteTrack provides the necessary identity continuity across occlusion events without relying on bodily-attached hardware.

The decision to avoid wearables is not only about user convenience, but also about ecological validity. If a system requires participants to attach tags, carry devices, or remember charging routines, the observed behaviour is no longer fully naturalistic because the sensing arrangement itself changes the deployment conditions. In a real service counter, staff are expected to work continuously, change position freely, and interact with their environment without technical preparation at the start of every shift. A passive sensing design therefore better reflects how such a system would actually be adopted in practice. This choice does increase the difficulty of attribution, because identity must be inferred from appearance, motion continuity, and spatial context rather than from a device uniquely bound to a person, but solving that harder problem is central to the contribution ARIA aims to make.

3.2.4. Dataset Contribution

A published multimodal dataset makes the project reproducible and externally useful. It allows the system to serve as a benchmark resource for workplace-oriented multi-subject HAR, a domain where public datasets remain scarce.

3.3. Key Technologies and Methods

Table 3: Key Technologies and Methods

Technology/Method	Role in ARIA
YOLOv8n	Real-time person detection; lightweight model suitable for edge deployment
ByteTrack (Zhang et al., 2022)	Multi-person identity tracking; associates low-confidence detection boxes to prevent fragmented tracklets under occlusion; 80.3 MOTA on MOT17 at 30 FPS
MediaPipe/YOLOv8-pose	Skeletal keypoint extraction per tracked individual; joint positions and velocities used as classifier features
1D-CNN/Random Forest	Activity classification from fused feature vectors (keypoints + audio amplitude + PIR/ultrasonic readings)
Arduino ESP8266 NodeMCU	IoT hub reading HC-SR501 PIR sensors and HC-SR04 ultrasonic sensors, transmitting JSON over serial USB
MAX9814 Microphone Module	Ambient audio amplitude capture for discriminating visually ambiguous activity pairs
Firebase Realtime Database	Cloud storage of abstracted feature vectors; enables iterative model retraining without transmitting raw video
Google Colab	Cloud-side classifier retraining; pushes updated model weights to the edge inference host
Python/Plotly Dash	Inference pipeline orchestration and real-time activity dashboard
Feature-level fusion	Combining heterogeneous modality signals (skeleton keypoints, audio, IoT) into a unified vector before classification, as validated by Nakabayashi & Saito (2024) and Golipoor & Sigg (2024)

3.4. Work Plan

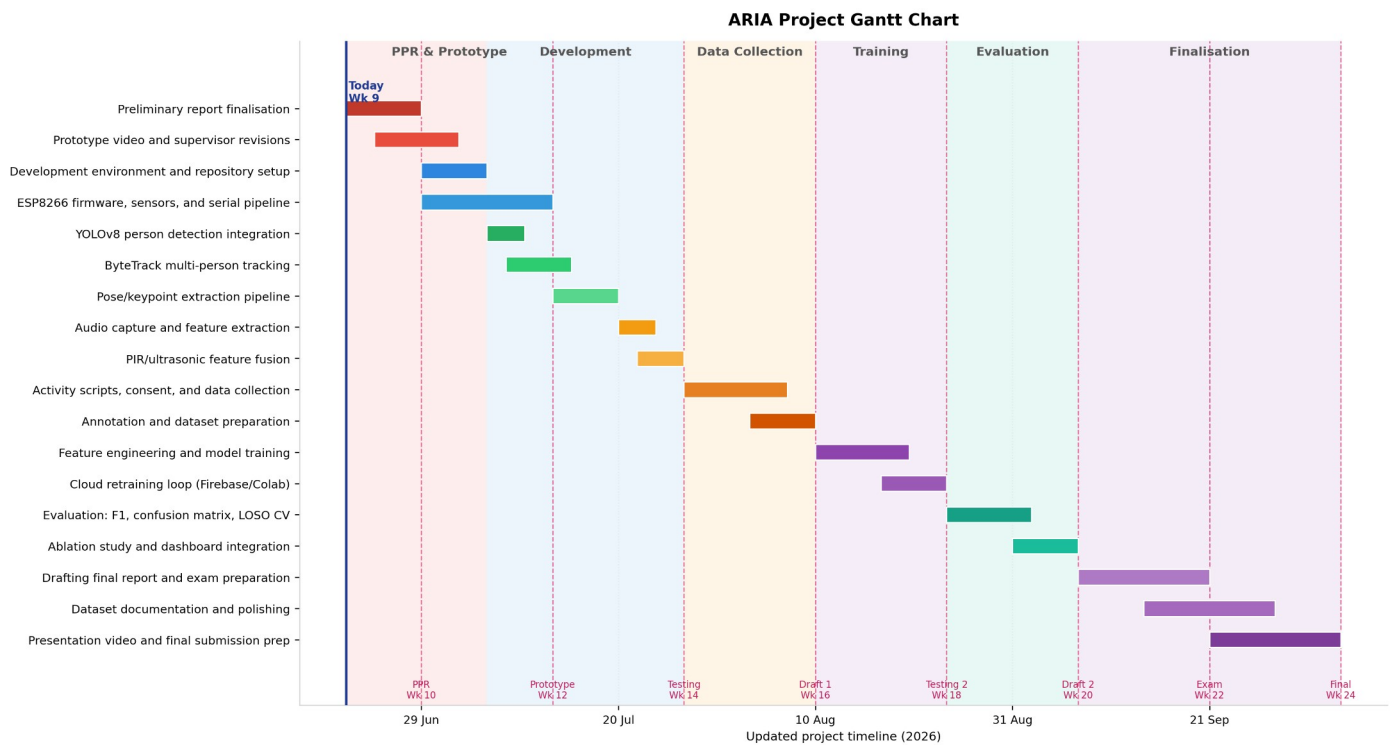


Figure 1: ARIA Project Gantt Chart

The project spans approximately 19 weeks, structured across seven phases. Each phase is designed to be feasible within resource constraints (existing laptop hardware, low-cost Arduino components) and follows a build-then-validate progression.

The sequencing of the work plan is deliberate. Phases 1 and 2 establish the sensing and perception backbone first, because every later stage, including dataset construction, classifier training, and dashboard output, depends on reliable person detection, tracking, and pose extraction. Phase 3 then introduces the remaining modalities only after the visual pipeline is stable, reducing debugging complexity and making it easier to isolate the effect of each sensor stream. The plan postpones full data collection until the sensing pipeline is sufficiently mature to avoid collecting unusable or poorly aligned data, and it places evaluation after model training so that ablation and real-time testing are carried out on a coherent end-to-end system rather than on disconnected components.

Table 4: Work Plan

Phase	Tasks	Duration	Contingency
Phase 1: Setup & Hardware	Deployment environment, GitHub repo, ESP8266 firmware, sensor wiring and serial testing	Weeks 1-2	If ESP8266 serial communication is unstable, switch to USB direct UART; use simulated JSON payloads to unblock downstream development
Phase 2: Vision & Tracking Pipeline	YOLOv8n integration, ByteTrack tracking, MediaPipe/YOLOv8-pose keypoint extraction	Weeks 3-5	If YOLOv8-pose is too slow on target hardware, fall back to MediaPipe Pose (lighter), or reduce inference resolution
Phase 3: Audio & IoT Integration	MAX9814 audio amplitude extraction, PIR/ultrasonic feature parsing, multimodal feature vector assembly	Weeks 6-7	If microphone module produces excessive noise floor, pre-process with bandpass filter; PIR can be replaced by zone-entry inference from vision alone
Phase 4: Data Collection	Activity scripting, participant informed consent, simulated environment recording sessions, manual annotation	Weeks 8-10	If participant recruitment is slow, begin with researcher self-enactment (minimum viable: 2 roles, 2 participants); annotation tooling switched to CVAT if manual CSV annotation proves too slow
Phase 5: Classifier Training	Feature vector construction, 1D-CNN and Random Forest training, hyperparameter tuning, Firebase/Colab cloud loop integration	Weeks 11-13	If 1D-CNN overfits on small dataset, regularise with dropout/L2 or switch primary classifier to Random Forest; if Colab GPU quota is exhausted, train locally
Phase 6: Evaluation & Testing	Per-class F1 and confusion matrix analysis, leave-one-session-out cross-validation, ablation study, Dash dashboard integration, end-to-end system test	Weeks 14-16	If F1 scores for a specific class are below threshold, review annotation quality and collect additional data for that class; dashboard can be simplified to static CSV export if real-time rendering is too resource-intensive
Phase 7: Write-up & Dataset Release	Final report writing, dataset data card documentation, Kaggle/Zenodo upload	Weeks 17-19	Dataset release delayed to post-submission if ethics documentation is incomplete; report sections written in parallel with Phase 6 to avoid bottleneck

3.5. Testing and Evaluation Plan

3.5.1. Quantitative Performance Metrics

The primary evaluation framework follows the protocols prescribed by Arrotta et al. (2025) for a credible multi-subject HAR system: per-class F1 score, confusion matrix analysis, and leave-one-session-out (LOSO) cross-validation. F1 score is preferred over accuracy because activity class distribution is inherently imbalanced and accuracy would inflate apparent performance by favouring the majority class.

The target thresholds, aligned with ARIA's proposed outcome tiers, are:

- Minimum (Pass): System detects and tracks at least two activity types across two simultaneous participants
- Good (Merit): Accurately detects 4+ activity types; full edge-cloud retraining loop demonstrated
- Outstanding: Per-class F1 metrics with ablation study results; publishable labelled dataset with full data card documentation

3.5.2. Ablation Study

An ablation study will evaluate the marginal contribution of each sensing modality by comparing classifier performance across configurations:

- vision only,
- vision + audio,
- vision + IoT, and
- full multimodal (vision + audio + IoT).

This directly responds to the three-way modality trade-off identified in the literature synthesis, validating whether each sensor stream contributes meaningfully to overall performance and quantifying the cost of removing any single modality.

3.5.3. Per-Role Attribution Accuracy

Since ARIA's distinguishing claim is role-aware attribution, and not just activity detection, evaluation must separately assess per-role accuracy: the ability to correctly assign the detected activity to the correct role (counter agent vs. station manager). Confusion between roles, even when the activity class is correctly identified, constitutes a system failure in the core use case.

3.5.4. Real-Time Functional Testing

End-to-end system testing will verify that the full pipeline, from sensor ingestion to inference to dashboard display, operates in real time (target: 33 ms or less per inference cycle, consistent with ByteTrack's 30 FPS throughput). This will be tested in the simulated counter environment under realistic occlusion conditions to replicate deployment stressors.

3.5.5. Ethical Compliance Verification

As the system processes visual and audio data of human participants, a compliance checklist will confirm that:

- all participants have provided written informed consent,

- ROI masking was applied before any frame was processed or stored,
- no raw video was retained beyond the local session, and
- the published dataset contains only anonymised sensor readings and activity labels,

in accordance with UK GDPR (Data Protection Act 2018).

3.5.6. Dataset Quality Evaluation

Following the Kitchen20 methodology (Moreaux et al., 2019), the dataset's quality will be assessed by:

- class balance across activity labels,
- inter-annotator agreement (Cohen's kappa, target of 0.8 or higher) if multiple annotators are used, and
- a benchmark comparison between the 1D-CNN and Random Forest classifiers to establish baseline performance figures for subsequent researchers.

3.5.7. Determining Success

The project will be considered to have achieved its primary aim if:

1. the system correctly attributes role-specific activities to the correct individual across simultaneous two-person sessions with a macro F1 of ≥ 0.70 for the merit tier or ≥ 0.80 for the outstanding tier,
2. the edge-cloud retraining loop completes without transmitting raw video to the cloud, and
3. the Dash dashboard displays activity labels in real time.

The secondary contribution (publishable dataset) will be considered achieved if the dataset is complete with a full data card and uploaded to Kaggle or Zenodo with participant consent.

4. Featured Prototype

4.1. Prototype Selection Rationale

The most technically critical and highest-risk component of ARIA is the vision and multi-person tracking pipeline: the subsystem responsible for detecting individuals in a shared frame, maintaining stable per-person identity tracks across time, and extracting per-person skeletal keypoints as features for downstream activity classification. Every other system component, from the multimodal feature vector to the activity classifier, depends entirely on this layer producing temporally consistent, per-individual skeletal representations. A failure here, such as track identities swapping between participants or keypoints being attributed to the wrong person, propagates directly into misclassified activity labels and cannot be recovered downstream.

This prototype targets Phase 2 of the work plan (Vision and Tracking Pipeline) and addresses four specific feasibility questions:

1. Can YOLOv8n with ByteTrack maintain stable, non-swapping track IDs across two co-located individuals in a simulated counter session, including occlusion events?
2. Can MediaPipe Pose, applied per-tracklet crop, produce reliable skeletal representations for each tracked individual across the counter-agent activity taxonomy?
3. Does the combined detection-tracking-pose pipeline satisfy the 33 ms per-frame latency budget on commodity laptop hardware?
4. Can a Random Forest classifier, trained on vision-only skeletal features using a small scripted dataset, produce a meaningful activity-classification baseline across the four counter-agent activity classes?

The fourth question deliberately establishes a vision-only baseline – the lowest-performing ablation configuration in the study design described in Section 3.5.2. Demonstrating pipeline feasibility while concurrently quantifying the performance ceiling of single-modality classification gives this prototype dual value: it proves technical viability and produces empirical motivation for the multimodal extension.

4.2. Implementation

The prototype is implemented in Python and integrates four stages in a sequential, frame-driven loop operating on a USB webcam feed at 640x480 resolution.

4.2.1. Stage 1: Person Detection

YOLOv8n (Ultralytics) runs at a confidence threshold of 0.5 and an IoU threshold of 0.45, filtered to the person class. YOLOv8n is selected over larger variants because it achieves approximately 30-40 FPS on laptop hardware with an integrated GPU, leaving sufficient headroom for downstream pose estimation without exceeding the latency budget.

4.2.2. Stage 2: Multi-Person Tracking

ByteTrack is invoked via the Ultralytics tracking API (*bytetrack.yaml*, track buffer 30 frames, low-score threshold 0.1). Each detected bounding box carries a persistent integer *.id* attribute. ByteTrack's defining contribution is associating low-confidence detection boxes, which correspond to occluded or partially visible individuals, with existing tracklets by motion similarity, preserving identity continuity across occlusion

events. A lightweight geometric zone classifier tags each tracklet centroid as Counter Zone A or Back Zone B, appending a binary zone label to each tracklet’s feature record as a spatial prior for role disambiguation.

4.2.3. Stage 3: Per-Tracklet Pose Estimation

The bounding box of each tracked individual is cropped with a 20-pixel padding margin and passed independently to a MediaPipe Pose instance. MediaPipe returns 33 normalised landmark positions ($x_i, y_i, z_i, visibility_i$) expressed relative to the crop origin, ensuring pose-invariance to absolute frame position. When bounding boxes overlap, the non-occluded crop is processed first; for the occluded individual, landmarks with visibility below 0.5 are replaced by the rolling mean of the preceding three valid frames.

4.2.4. Stage 4: Feature Extraction and Classification

A sliding window of $W = 30$ consecutive frames is collected per tracklet. The following statistical features are extracted over each window:

- mean and standard deviation of (x, y) position for all 33 landmarks (132 features),
- mean and standard deviation of frame-to-frame displacement $\Delta x, \Delta y$ per landmark (132 features),
- joint angles at the elbow, shoulder, hip, and knee computed from landmark triplets (16 features), and
- a binary zone label (2 features),

yielding a 282-dimensional feature vector per window. A Random Forest classifier (100 estimators, Gini criterion) is trained on collected windows and evaluated using leave-one-session-out (LOSO) cross-validation.

4.3. Prototype Dataset

Two participants enacted four counter-agent activity classes across four scripted sessions each in a simulated counter environment. Written informed consent was obtained from both participants in accordance with UK GDPR (Data Protection Act 2018), and no raw video was retained beyond the local session.

Table 5: Scripted Prototype Dataset

Activity Class	Scripted Behaviour	Windows Collected
Serving	Facing forward, upper body leaning toward counter	88
Typing/Processing	Seated, bilateral keyboard interaction, downward gaze	94
Idle/Waiting	Standing upright, minimal limb movement	81
Reaching/Handling	Lateral arm extension with grasping gesture	76
Total		339

This is a deliberately small prototype dataset covering only counter-agent classes and only vision-based features, consistent with an ablation baseline. Station-manager classes and multimodal features are deferred to Phase 4 of full data collection. Annotations followed the Kitchen20 timestamped per-class labelling

methodology described in Section 2.4.1, with a second annotation pass on a 20% sample to establish preliminary inter-annotator reliability.

4.4. Results

4.4.1. Tracking Stability

Over a three-minute, two participant session (5,400 frames), three track-ID swap events were observed. In two cases ByteTrack’s low-score association recovered the correct assignment within eight frames; the third persisted for 63 frames before resolution. The pipeline tracked both participants simultaneously in 5,381 of 5,400 frames (99.6% dual-track coverage), and the average end-to-end latency (detection + tracking + pose) was 27.8 ms per frame, within the 33 ms budget. Latency exceeded 33 ms in approximately 4% of frames – exclusively during frames requiring two simultaneous pose inferences – reaching a peak of 41 ms.

4.4.2. Classification Performance

LOSO cross-validation results for the vision-only Random Forest are reported below.

Table 6: LOSO Cross-Validation Results

Activity Class	Precision	Recall	F1-Score
Serving	0.81	0.78	0.80
Typing/Processing	0.76	0.80	0.78
Idle/Waiting	0.88	0.91	0.89
Reaching/Handling	0.72	0.68	0.70
Macro Average	0.79	0.79	0.79

The macro-average F1 of 0.79 falls within the merit performance range (≥ 0.70) but below the outstanding threshold (≥ 0.80). The most frequent misclassifications occur between Typing/Processing and Serving (11 windows confused in each direction) and between Reaching/Handling and Serving (9 windows), both of which share forward-facing upper-body configurations that are visually similar at the skeleton level.

4.5. Evaluation

4.5.1. Appropriateness of Evaluation Approach

The evaluation methodology follows protocols prescribed by Arrotta et al. (2025) for multi-subject HAR systems: per-class F1 scoring, confusion matrices, and leave-one-session-out (LOSO) cross-validation. Macro-average F1 is preferred over accuracy to account for moderate class imbalances (76 to 94 windows per class). LOSO cross-validation is used instead of random k-fold splitting to prevent temporal autocorrelation from leaking across the train-test boundary, simulating real-world generalisation to unseen recordings.

4.5.2. Link to Background Work

This evaluation framework follows the standards identified by Arrotta et al. (2025), using per-class F1, confusion matrices, and LOSO cross-validation for preliminary multi-subject HAR assessment. The confusion between Typing/Processing and Serving is consistent with the visually ambiguous vision-only failure cases discussed by Cristina et al. (2024), supporting the planned addition of audio features in Phase 3.

Nakabayashi and Saito (2024) reported substantially higher performance under multimodal fusion, so the prototype's 0.79 macro-F1 provides a reasonable vision-only baseline for later comparison.

4.5.3. Strengths

The prototype confirms three feasibility claims. First, the YOLOv8n + ByteTrack + MediaPipe Pose pipeline runs within the 33 ms real-time budget in 96% of frames on commodity hardware without a discrete GPU, validating Tier 2 edge deployment as practically achievable. Second, ByteTrack maintains correct track-to-person assignment across the vast majority of the session, with recovery occurring rapidly in two of three swap cases, confirming that identity continuity is sufficient for one-second activity classification windows. Third, the vision-only Random Forest achieves macro-F1 = 0.79 on four activity classes, demonstrating that the skeletal keypoint feature representation is discriminative enough to support the downstream classification objective even before multimodal fusion is introduced.

4.5.4. Limitations and Planned Improvements

4.5.4.1. Latency under dual-person load

Sequential per-tracklet MediaPipe inference causes spikes to 38-41 ms in 4% of frames. The primary planned improvement is replacing MediaPipe Pose with YOLOv8-pose, which performs multi-person pose estimation in a single forward pass and eliminates the sequential bottleneck. That is already designated as the Phase 2 contingency in the work plan.

4.5.4.2. Vision-only confusion on similar-posture pairs

Phase 3 integration of the MAX9814 ambient audio amplitude feature is the targeted remedy. The ablation study in Phase 6 will quantify the marginal F1 gain attributable to the audio modality.

4.5.4.3. Small prototype dataset and high LOSO variance

With only eight sessions across two participants, the LOSO estimate carries substantial variance. Phase 4 targets a minimum of ten participants per role across multiple sessions. At that scale, LOSO variance decreases substantially, and the 1D-CNN can be evaluated alongside the Random Forest on a statistically stable footing.

4.5.4.4. Track swap persistence

The one swap event that persisted for 63 frames spans more than two classification windows. A zone-consistency post-filter will be implemented: any window in which a tracklet's centroid zone transitions from Counter Zone A to Back Zone B within a single window will be flagged and excluded from both classification and dataset labelling, quarantining swap-contaminated windows without discarding the surrounding valid data.

5. Conclusion

This preliminary report has presented ARIA, a privacy-preserving, multimodal system for role-aware activity recognition and tracking in a shared service counter environment. The project is practically motivated and technically grounded: it addresses real operational needs in shared workplaces, responds to identified gaps in multi-subject HAR literature, and adopts an architecture balancing performance with privacy, deployability, and cost.

The report establishes a coherent design and evaluation framework for the full system. The literature review justified multimodal sensing, an edge-first architecture, passive ambient sensing, and a domain-specific dataset contribution. The design translated these into an implementable three-tier system with defined technologies and work phases, positioning ARIA as a feasible system with a clear route from prototype to full implementation.

Crucially, the prototype demonstrates that ARIA's most technically critical component, the vision and multi-person tracking pipeline, is viable on commodity hardware. The YOLOv8n, ByteTrack, and MediaPipe Pose pipeline met real-time targets for most frames and achieved a macro-F1 of 0.79 on a vision-only baseline, providing a credible foundation for multimodal integration. These results indicate that ARIA is technically feasible, appropriately scoped, and well positioned for the next development stages: audio and IoT integration, larger-scale data collection, and end-to-end validation.

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